

PAPER_1552 — Recombination Redshift = $z_{\text{recomb}} = A_5 \cdot SO_5 + A_5 \cdot D_{\text{phys}} + SO_5 \cdot D_{\text{crit}} - SO_5 = 600 + 240 + 260 - 10 = 1090$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Cosmology **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure from PAPER_1209GG_S651

Observation

PAPER_1209GG_S651 (Cosmology Unified Proof Set) lists **Recombination Redshift** with the integer-primitive expression:

Recombination Redshift = $z_{\text{recomb}} = A_5 \cdot SO_5 + A_5 \cdot D_{\text{phys}} + SO_5 \cdot D_{\text{crit}} - SO_5 = 600 + 240 + 260 - 10$
Status: **EXACT**.

UQFF Closed Identity

$z_{\text{recomb}} = A_5 \cdot SO_5 + A_5 \cdot D_{\text{phys}} + SO_5 \cdot D_{\text{crit}} - SO_5 = 600 + 240 + 260 - 10 = 1090$ EXACT

Physical Interpretation

z_{recomb} sets the last-scattering surface of the cosmic microwave background — the epoch (~380,000 yr after Big Bang) when electrons first bound to nuclei and photons decoupled.

Four integer products with zero fitted constants: - $A_5 \cdot SO_5 = 600$ (icosahedral $\times$ decimal scale) - $A_5 \cdot D_{\text{phys}} = 240$ (icosahedral $\times$ spacetime dimensions) - $SO_5 \cdot D_{\text{crit}} = 260$ (decimal $\times$ bosonic-string critical dimension) - $-SO_5 = -10$ (decimal correction)

Sum = 1090 EXACT — the Planck-2018 best-fit value for z_{recomb} .

This is one of the cleanest cosmological closures in UQFF: the recombination epoch derives from integer-primitive arithmetic alone.

NOT REPLACEMENT

CODATA/PDG treats Recombination Redshift as a precision-measured constant. UQFF supplies the structural integer-primitive form, demonstrating the framework's predictive economy without claiming to replace experimental measurement.

Reference

- Source: PAPER_1209GG_S651
- Related: PAPER_1209 series unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "z_recomb_1090"})`

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